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Review article

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Advances in the treatment of ANCA-associated vasculitis

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Abstract

Antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis (AAV) consists of a group of small-vessel vasculitides that often present with organ-threatening or life-threatening manifestations. Current immunosuppressive treatments have improved survival and rates of remission, but are not curative, have frequent toxicities, and do not effectively prevent relapse. Clinical trials have established the role of rituximab, an anti-CD20 B cell-depleting monoclonal antibody, in both the remission-induction and maintenance phases of the disease and demonstrated that glucocorticoid doses can be substantially reduced from historical dosing levels without affecting treatment efficacy. Therapies that have the potential to be more effective and safer have become available or are under investigation. Avacopan, an oral C5a receptor antagonist, was approved as an adjunctive treatment for AAV and use of this drug in combination with rituximab or cyclophosphamide and markedly reduced glucocorticoid dosing demonstrated superior efficacy and potentially greater kidney recovery than prior standard of care. Other agents under study for treatment of AAV include next-generation anti-CD20 monoclonal antibodies, anti-CD19 chimeric antigen receptor T cells, novel complement inhibitors and agents that can target fibrosis. Alongside traditional randomized controlled trials with clinical endpoints, experimental medicine studies are focusing on mechanistic endpoints and disease biomarkers. This Review discusses current treatments and the advances in the management of AAV.

Sections

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Key points

- The main regimen of remission induction in antineutrophil cytoplasmic antibody-associated vasculitis is based on high-dose glucocorticoids combined with rituximab or cyclophosphamide.
- The initial approach to treatment induces remission of vasculitis in most patients but does not prevent relapse in a substantial number of people and has multiple toxicities.
- Clinical trial data show that reduced-dose regimens of glucocorticoids are non-inferior to standard-dose regimens and have a lower risk of serious infections; rituximab is superior to azathioprine in maintaining disease remission.
- B cell-targeting therapies under evaluation for antineutrophil cytoplasmic antibody-associated vasculitis treatment include anti-CD20 monoclonal antibodies with increased B cell depletion capacity compared with rituximab, chimeric antigen receptor T cells and B cell-inhibiting antibodies.
- Avacopan in combination with rituximab or cyclophosphamide and markedly reduced glucocorticoids has superior efficacy, kidney function recovery and improvement in quality of life to regimens based on standard glucocorticoid dose.
- Novel targets include pathways involved in antineutrophil cytoplasmic antibody-induced inflammation and fibrosis, which could be combined with immunosuppressive therapies. Strategies to reduce cardiovascular risk and improve response to vaccines are being tested.

Introduction

Antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis (AAV) encompasses three rare disorders – microscopic polyangiitis (MPA), granulomatosis with polyangiitis (GPA) and eosinophilic granulomatosis with polyangiitis (EGPA) – which are characterized by inflammation of small blood vessels and scarce immune deposition[1]. MPA and GPA show high frequencies of necrotizing crescentic glomerulonephritis and alveolar haemorrhage that result from small-vessel vasculitis and are usually associated with ANCA positivity. Furthermore, a key component of GPA but not MPA is granulomatous inflammation, which mainly affects the ears, nose and throat (ENT) and lungs[2]. The main features of EGPA are asthma and eosinophilia; glomerulonephritis is rare and only 30–40% of patients test positive for ANCA[3,4]. The treatment of EGPA, which differs substantially from that of GPA and MPA, is not discussed in this Review and evidence-based guidelines for the management of this disorder were published in 2023 (ref. 5).

In individuals with MPA and GPA, ANCAs mainly target myeloperoxidase (MPO) or proteinase 3 (PR3), which are primarily found in neutrophils, but also in monocytes and macrophages. Antigen specificity is associated with clinical phenotype and prognosis[6]. PR3-ANCA is more common in GPA and is associated with younger age at onset, higher rates of relapse and lower mortality, whereas MPO-ANCA is prevalent in MPA and portends a higher risk of kidney involvement, progression to end-stage kidney disease (ESKD) and death[7–11]. Genetic associations align more with the type of ANCA than with clinical diagnosis[12,13]; hence, distinguishing between PR3-AAV and MPO-AAV, as opposed to GPA and MPA, could improve risk stratification of patients and reflect more accurately pathogenic differences[14,15]. However, even such an ANCA-based classification fails to separate AAV into two distinct entities as both show overlapping features and substantial heterogeneity. PR3-AAV can be associated with localized ENT disease or with systemic presentation dominated by kidney involvement indistinguishable from that seen in MPO-AAV, whereas some patients classified as GPA are positive for MPO-ANCA; this finding is mainly reported in regions such as China and Japan, where MPO-ANCA is the dominant serotype[16,17]. Model-based clustering of large patient datasets have identified multiple disease clusters, such as non-renal AAV, renal AAV with or without PR3-ANCA and smaller clusters that are characterized by gastrointestinal and central nervous system involvement, thereby providing evidence that a binary classification will not resolve the complexity of the AAV spectrum[18–20]. Defining patient subgroups based on the presence of granulomatous features or kidney involvement, which confer an increased risk of relapse and death, respectively, has also been proposed[21].

Although the aetiology of AAV remains unknown in most cases, the pathogenesis of vasculitic lesions has been delineated more clearly[22]. Autoreactive B cells, assisted by T cells, give rise to ANCA-producing plasmablasts and plasma cells. Circulating ANCAs bind to cell-surface MPO and PR3 expressed by neutrophils and monocytes, which, in the presence of pro-inflammatory cytokines and complement factors, causes neutrophil activation, degranulation and release of neutrophil extracellular traps (NETs). The subsequent tissue inflammation ultimately results in fibrosis[23]; however, although ANCAs have an established pathogenetic role in AAV, other factors also contribute. Notably, individuals with active disease might be ANCA-negative and others that do not exhibit any features of AAV can test positive for ANCA[24]. Multiple MPO and PR3 epitopes, as well as alternative ANCA specificities, could account for differences in pathogenicity[25,26].

The use of standard immunosuppressive therapies, such as cyclophosphamide, has improved survival in patients with AAV and converted a frequently fatal condition to one that follows a chronic relapsing–remitting course[27,28]. However, a substantial proportion of patients still develop end-organ damage, such as ESKD, experience disease relapse or suffer toxicities from treatment, which affects quality of life and life expectancy[29,30]. The identification of key pathogenetic players, including B cells and the complement system, supports the development of so-called ‘targeted’ therapies, such as rituximab and avacopan, and fosters the evaluation of innovative treatments that could expand therapeutic options for AAV. Furthermore, strategies to reduce the occurrence of infections, a leading cause of death among patients with AAV, and to address comorbidities that often accompany vasculitis, such as chronic kidney disease (CKD), are being investigated or are already available. This Review discusses current treatment, advances and controversies in the management of AAV.

Current treatment and controversies

Patients with MPA and GPA are often grouped together in clinical trials and receive similar immunosuppressive regimens, which typically includes a 3–6-month phase of ‘induction of remission’ followed by a prolonged phase of ‘maintenance of remission’. Remission is defined as the cessation of disease activity, and should be qualified by time since treatment start, and the need for ongoing therapy, such as oral glucocorticoid dose[31]. These parameters have varied between trials, meaning that remission rates cannot be directly compared between studies, although two major trials have used a Birmingham Vasculitis

Activity Score of zero at 6 months and glucocorticoid withdrawal in their remission definitions[32,33]. As biomarkers for active disease and remission are scarce, disease activity is quantified using the validated Birmingham Vasculitis Activity Score, or its derivatives, a checklist of clinical features of active disease, such as haematuria and nasal crusting[34]. Certain clinical features are not unique to vasculitis and thus distinguishing those symptoms that occur as a result of other causes, including damage (that is, irreversible organ dysfunction no longer responsive to immunosuppression), can be problematic. Long-term sequelae of disease and/or treatment also include infertility, malignancy and cardiovascular disease.

Guidelines for the management of AAV have been updated by international societies in the past 5 years[35–37]. The following sections discuss the induction and maintenance of remission, with Table 1 and Table 2 outlining features of the main therapies. The main controversies on the use of current treatments are summarized in Table 3.

Induction of remission

The main regimen to induce remission in AAV is a combination of glucocorticoids with rituximab or cyclophosphamide. This approach results in remission within 6 months in 70–90% of patients, depending on the stringency of the definition, the threshold dose of glucocorticoids applied and the other immunosuppressive therapies used[36]. For patients without organ-threatening or life-threatening manifestations, such as glomerulonephritis, alveolar haemorrhage, mononeuritis multiplex, meningeal involvement and retro-orbital disease[36,38], alternatives to cyclophosphamide and rituximab include mycophenolate mofetil and methotrexate but use of these drugs is complicated by higher relapse rates than cyclophosphamide[39–42]. Furthermore, plasma exchange (PLEX) and intravenous immunoglobin are used in addition to immunosuppressive therapies for selected patients. In 2021, avacopan, an oral complement inhibitor, was approved as an adjunctive treatment to cyclophosphamide or rituximab and a lower than standard dosing regimen of glucocorticoids.

Refractory disease, that is, failure to achieve remission with standard treatment protocols, inability to taper glucocorticoids to a low dose or disease progression, occurs in 10–30% of patients with AAV[31]. Treatment options include switching from rituximab to cyclophosphamide and vice versa, adding avacopan, PLEX or intravenous immunoglobin, but the prognosis and quality of life remain poor in this group[31,43]. Causes of secondary vasculitis, including recreational drugs and malignancy, should be considered in all patients, particularly those with refractory disease[44].

Glucocorticoids. Findings from clinical trials indicate that the dose of glucocorticoids, a major determinant of toxicity, can be substantially reduced from the previous standard dosing without affecting treatment efficacy (Table 1). In the international PEXIVAS trial of over 700 patients with AAV with kidney involvement (estimated glomerular filtration rate (eGFR) < 50 ml/min/1.73 m^2) and/or alveolar haemorrhage treated with cyclophosphamide or rituximab, a reduced glucocorticoid tapering regimen, resulting in a 40% lower cumulative dose of oral prednisone at 6 months than the standard tapering regimen, was non-inferior for efficacy and associated with fewer serious infections[45]. The LoVAS trial also found reduced-dose prednisolone (0.5 mg/kg per day) in combination with rituximab to be non-inferior to high-dose prednisolone (1 mg/kg per day). However, in this study, all of the patients were from Japan and had predominantly newly diagnosed AAV with MPO-ANCA and mild kidney involvement, potentially limiting the generalizability of the results[46]. In the international RITAZAREM trial, which enrolled patients with relapsing disease, investigators were allowed to choose between prednisone starting at 0.5 or 1 mg/kg per day, alongside rituximab, and no difference in remission rates was observed between the two glucocorticoid regimens[47]. Overall, these results encourage minimization of the dose of glucocorticoids, but these drugs remain a highly valuable treatment in the initial management of AAV. High-dose intravenous glucocorticoids, usually 1–3 g of pulse methylprednisolone over 3 days, are often administered prior to commencing oral prednisolone, and their use was allowed in most trials, including those investigating oral glucocorticoid reduction[33,45,47,48]. No randomized trial has assessed whether the addition of methylprednisolone pulses is beneficial in AAV; hence, evidence is lacking to support this practice (Table 3). Some studies report no difference in efficacy but an increased risk of infections with intravenous glucocorticoids, and their use is recommended only for patients with features of organ-threatening or life-threatening disease, such as those in the PEXIVAS trial[36,49,50].

Rituximab and cyclophosphamide. Rituximab, a B cell-depleting, chimeric monoclonal antibody that targets CD20, has been increasingly used since the RAVE study found it to be non-inferior for remission induction to cyclophosphamide, a cytotoxic agent that has long been standard of care in AAV, followed by azathioprine[32]. Compared with cyclophosphamide, rituximab is not associated with an increased risk of malignancy or infertility, and was superior for those individuals with relapsing disease and PR3-ANCA positivity[51,52] (Table 1). Furthermore, a post hoc analysis of the same trial showed no difference in kidney outcomes between the two regimens[53].

However, because those with creatinine >4 g/dl were excluded from the RAVE trial, it remains unknown whether rituximab and cyclophosphamide are also equivalent in patients with a substantially reduced eGFR. Retrospective observations show similar results of both regimens among those with advanced kidney failure and in the PEXIVAS trial the type of induction therapy was not associated with the composite outcome, although the trial was not powered to compare rituximab and cyclophosphamide[45,54–56]. Notably, the current recommended PEXIVAS reduced-dose glucocorticoid schedule is lower than the oral glucocorticoid regimen used in the RAVE trial; however, most patients in the PEXIVAS trial received cyclophosphamide and among those treated with rituximab there was a trend favouring the standard-dose glucocorticoid regimen, which raises concern that reduced-dose glucocorticoids might be less effective than the standard dose when given with rituximab[45]. Unlike cyclophosphamide, rituximab does not have a direct effect on myeloid cells that drive inflammation, such as neutrophils and monocytes, which might result in a slower control of disease manifestations[57,58].

Another option is the combined use of rituximab and cyclophosphamide (Table 3). The RITUXVAS trial showed that rituximab could be cyclophosphamide sparing among patients with severe renal AAV, including those with dialysis dependence. The rituximab treatment group only had 2–3 doses of intravenous cyclophosphamide, compared with 6–10 doses in the standard cyclophosphamide-only group, with no difference in the frequency of adverse events[59]. The combination of rituximab and low-dose cyclophosphamide has also been shown in observational studies to be potentially glucocorticoid sparing[60]. A randomized trial is currently underway to test whether the addition of cyclophosphamide pulses to rituximab enables a more sustained remission compared with standard rituximab therapy (NCT03942887).

Table 1 | Main therapies used for induction of remission in antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis (AAV)

Therapy	Mechanism	Usual dosing	Use	Adverse effects	Key studies	Considerations
Glucocorticoids	Binding of the glucocorticoid receptor inhibits innate and adaptive immune cells	Intravenous: 1–3 g[a] Oral: 50–70 mg per day or 0.5 mg/kg per day[b]	Almost all patients will receive oral glucocorticoids in combination with immunosuppressive agents; intravenous use is not established but is often used in organ-threatening or life-threatening disease[e]	Infections, diabetes mellitus, hypertension, obesity, osteoporosis, cataracts and psychosis	PEXIVAS[45] LoVAS[46] RITAZAREM[47]	Rapid tapering to 5 mg per day over 3 months is effective and associated with fewer serious infections than standard tapering; use of lower doses is possible for selected patients, especially with avacopan
Rituximab	Depletion of CD20[+] B cells via ADCC, CDC or DCD	Intravenous: 1 g on days 0 and 14 or 375 mg/m^2 weekly for 4 weeks	All patients	Infusion reactions, infections, hypogammaglobulinaemia, poor vaccine responses, late-onset neutropenia and PML (rare)	RAVE[32] RITUXVAS[59]	Superior to cyclophosphamide in PR3-AAV and relapsing disease but not formally compared in most severe kidney presentations[c]
Cyclophosphamide	Alkylating agent that suppresses the proliferation of innate and adaptive immune cells	Intravenous: 15 mg/kg[d] every 2–3 weeks for 6–10 doses Oral: 2 mg/kg per day[d] for 3–6 months	Administered for organ-threatening or life-threatening disease[e] if rituximab is not available or in combination with rituximab	Bone marrow suppression, infections, cystitis, sterility and malignancy (skin, bladder and bone marrow)	CYCLOPS[200,201] RITUXVAS[59]	Intravenous route is preferred (~50% lower cumulative dose); the addition of rituximab could be dose-sparing
Avacopan	C5aR antagonist that inhibits neutrophil chemotaxis, priming and activation	Oral: 30 mg twice daily for 1 year	Adjunctive therapy in patients at risk of glucocorticoid-related adverse effects, organ-threatening or life-threatening disease[e]	Liver toxicity (rare) Leukopenia (rare)	ADVOCATE[33]	Large glucocorticoid-sparing effect that enables quick tapering; kidney function recovery is greatest with poor baseline eGFR
Plasma exchange	Removal of circulating ANCA	Seven exchanges (60 ml/kg)	Adjunctive therapy for patients with kidney involvement and creatinine >300 µmol/l (3.4 mg/dl)	Increased risk of serious infections and bleeding and infusion reactions (uncommon)	PEXIVAS[45] MEPEX[48] Walsh et al.[64]	No effect on mortality; possible lower risk of ESKD at 1 year but also more infections; recommended for individuals with double-positive ANCA and anti-GBM antibodies
Mycophenolate mofetil	Inhibition of de novo purine synthesis, which suppresses lymphocyte proliferation	Oral: 1–1.5 g twice daily	Used as an alternative to rituximab in new-onset, non-organ-threatening[f] AAV	Poor gastrointestinal tolerance, bone marrow suppression and herpes zoster	MYCYC[39]	Higher relapse rate than cyclophosphamide in PR3-AAV
Methotrexate	Increased adenosine release, NOS uncoupling and increased sensitivity of T cells to apoptosis	20–25 mg per week oral or subcutaneous	Alternative to rituximab in new-onset, non-organ-threatening AAV[f]; contraindicated in eGFR < 30 ml/min/1.73 m^2	Nausea, diarrhoea, liver toxicity, leukopenia, chemical pneumonitis, liver fibrosis and dose adjustment needed with CKD	NORAM[42]	Less effective than cyclophosphamide in multiorgan and pulmonary disease
IVIG	Immunomodulating effects that are mediated by FcγR and FcRn	2 g/kg	Adjunctive therapy in refractory AAV (not recommended)	Adverse reactions (which are usually mild) and anaphylaxis in IgA deficiency	Jayne et al.[202]	Effects are not sustained in the long term; beneficial in patients with hypogammaglobulinaemia, high infection risk and PML

ADCC, antibody-dependent cytotoxicity; CDC, complement-dependent cytotoxicity; CKD, chronic kidney disease; DCD, direct cell death; eGFR, estimated glomerular filtration rate; ESKD, end-stage kidney disease; FcγR, factor crystallizable gamma receptors; FcRn, neonatal factor crystallizable receptor; GBM, glomerular basement membrane; IVIG, intravenous immunoglobulin; NOS, nitric oxide synthase; PML, progressive multifocal leukoencephalopathy; PR3, proteinase 3. [a]Methylprednisolone or equivalent. [b]Prednisone or equivalent. [c]Serum creatinine > 354 µmol/l (4 mg/dl) or alveolar haemorrhage requiring mechanical ventilation. [d]Dose adjusted to age and estimated glomerular filtration rate. [e]Glomerulonephritis, alveolar haemorrhage, mononeuritis multiplex, meningeal involvement, central nervous system involvement, retro-orbital disease, mesenteric involvement and cardiac involvement[36]. [f]ENT disease without bone involvement, cartilage collapse or deafness, skin involvement without ulceration, non-cavitating lung nodules and episcleritis[36].

Plasma exchange. PLEX is aimed at rapidly removing ANCAs and has been in use for 40 years as an adjunctive treatment for life-threatening manifestations of AAV (Table 1). The MEPEX trial reported a reduced risk of ESKD at 1 year for patients presenting with a serum creatinine >500 µmol/l (5.7 mg/dl) who received PLEX. However, results from the PEXIVAS trial, which was larger than the MEPEX trial, did not show a significant reduction in time to the composite outcome of death and/or ESKD over an average follow-up of 3 years for patients presenting with a eGFR <50 ml/min/1.73 m^2. There was a trend in the PEXIVAS trial that favoured PLEX on ESKD for the subgroup of patients with creatinine >500 µmol/l (5.7 mg/dl) but no effect on mortality was observed, which is perhaps explained by the multi-factorial nature of this end point. However, a post hoc analysis of the PEXIVAS trial found that early recovery of kidney function was significantly improved with PLEX and was associated with a reduced risk of ESKD at 12 months[61]. Furthermore, although PR3-ANCA is associated with more active histological lesions and greater kidney recovery than MPO-ANCA, eGFR improved with PLEX similarly in both groups, compared with individuals who did not receive PLEX, which indicates that the response to PLEX does not differ according to ANCA serotype[10,61–63].

An updated meta-analysis of nine randomized trials, including MEPEX and PEXIVAS, identified a significant reduction in the risk of ESKD, but not death, at 1 year in patients who received PLEX. Notably, the estimated absolute risk reduction was greater with increasing baseline creatinine, which was 5% for those with creatinine 300–499 µmol/l (3.4–5.6 mg/dl) and 16% for those with creatinine >500 µmol/l (5.7 mg/dl), suggesting that PLEX provides greater benefit in those at a higher risk of ESKD[64]. The meta-analysis also concluded that PLEX increases the risk of serious infections and benefits on kidney survival attenuate with time. As a result of these analyses, current guidelines and recommendations state that PLEX can be considered for patients with a serum creatinine >300 µmol/l[36,37,65] (Table 3). By contrast, PLEX should always be used in patients who are double-seropositive for ANCA and anti-glomerular basement membrane (GBM) antibodies; however, PLEX dosing in AAV is uncertain, with the probability that ANCA removal is incomplete, and no studies have used a PLEX regimen aimed at ANCA negativity, as is recommended for anti-GBM disease. Using more effective methods of reducing ANCA levels, such as immunoadsorption or imlifidase, will better address the question of the therapeutic benefit of this approach[66].

Therapies used for maintenance of remission

Relapse, which is defined as the reappearance of active manifestations following remission, occurs in one third of patients with AAV within 18 months of induction therapy[52,67]. Traditional disease-related risk factors include diagnosis of GPA, PR3-ANCA positivity, ENT involvement and previous history of relapse, but translating these risk factors into personalized strategies remains challenging[52,67,68]. In view of their toxicity, glucocorticoids should be reduced to low doses, such as 5 mg per day, by 6 months or earlier (Table 2). The effect of glucocorticoid discontinuation is unclear and has been investigated in patients with GPA who achieved remission by the TAPIR trial (NCT01933724). Preliminary results show that patients who discontinued treatment with prednisone had a higher frequency of 'minor' but not 'major' relapses

Table 2 | Main therapies used for maintenance of remission in antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis (AAV)

Therapy	Mechanism	Usual dosing	Indication	Adverse effects	Key studies	Considerations
Glucocorticoids	Binding of glucocorticoid receptor inhibits innate and adaptive immune cells	Oral: 5 mg per day[a] or less, discontinuation should be considered	Not always required	Obesity, diabetes mellitus, hypertension, osteoporosis, infections, cardiovascular disease, adrenal insufficiency, cataracts and psychosis	LoVAS[203] RITAZAREM[71] TAPIR[69]	Cumulative dose is associated with toxicity; discontinuation appears to be safe in patients on rituximab
Rituximab	Depletion of CD20$^+$ B cells via ADCC, CDC or DCD	Intravenous: 0.5–1 g every 6 months for 24 months or more	First choice for all patients	Infusion reactions, infections, hypogammaglobulinaemia, poor vaccine responses, late-onset neutropenia and PML (rare)	MAINRITSAN[70] RITAZAREM[71] MAINRITSAN-3[72]	Patients who frequently relapse might benefit from 4-monthly dosing and extended course (such as 1 g per year)
Azathioprine	Purine analogue that suppresses lymphocyte proliferation	Oral: 2 mg/kg per day orally for 24 months	Inferior alternative to rituximab; used after cyclophosphamide induction	Bone marrow suppression and liver toxicity	CYCAZAREM[204] REMAIN[73]	Prolonged course of >24 months is associated with fewer relapses; safe in pregnancy
Mycophenolate mofetil	Inhibition of de novo purine synthesis, which suppresses lymphocyte proliferation	Oral: 0.5–1 g twice daily for 24 months	Inferior to azathioprine in non-organ-threatening AAV[b]	Poor gastrointestinal tolerance, bone marrow suppression, herpes zoster and malignancy	IMPROVE[205]	Higher relapse rates and no safety advantage compared with azathioprine
Methotrexate	Increased adenosine release, NOS uncoupling and increased sensitivity of T cells to apoptosis	20–25 mg per week (oral or subcutaneous) for 24 months	Alternative to azathioprine in non-organ-threatening AAV[b]; contraindicated in eGFR <30 ml/min/1.73 m^2	Nausea, diarrhoea, liver toxicity, leukopenia, chemical pneumonitis, liver fibrosis, dose adjustment with CKD	WEGENT[206,207]	Similar rate of relapse to azathioprine but trend towards more adverse events
Leflunomide	Inhibition of uridine monophosphate synthesis and suppression of lymphocyte proliferation	Oral: 30 mg per day for 24 months	Alternative to azathioprine in non-organ-threatening AAV[b]	Hypertension, liver toxicity, leukopenia, rash, chemical pneumonitis and peripheral neuropathy	Metzler et al.[208]	Possibly superior to methotrexate but with a higher rate of adverse effects

ADCC, antibody-dependent cytotoxicity; CDC, complement-dependent cytotoxicity; CKD, chronic kidney disease; DCD, direct cell death; eGFR, estimated glomerular filtration rate; NOS, nitric oxide synthase; PML, progressive multifocal leukoencephalopathy. [a]Prednisone or equivalent. [b]Ear, nose and throat disease without bone involvement, cartilage collapse or deafness, skin involvement without ulceration, non-cavitating lung nodules and episcleritis[36].

Table 3 | Main controversies on current treatments for antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis (AAV) and potential solutions

Issue	Pro	Con	Considerations
Should treatment start with IV glucocorticoids?	Immediately available Possibly faster control of inflammation Used in several clinical trials	Benefit not assessed in clinical trials Substantial increase in cumulative glucocorticoid dose Possible higher risk of serious infections	Recommended only for organ-threatening or life-threatening presentations
Should a PEXIVAS reduced-dose regimen of oral glucocorticoids be used in patients with kidney involvement and/or alveolar haemorrhage?	Non-inferior to a standard-dose regimen in combination with cyclophosphamide followed by azathioprine or rituximab Reduced serious infections and GTI than standard-dose regimen	Trend favouring higher-dose regimen among patients on rituximab	Currently recommended Early use of avacopan could enable quicker and more effective tapering Caution in patients on rituximab only
Should low-dose cyclophosphamide be combined with rituximab in organ-threatening or life-threatening presentations (such as creatinine >300 µmol/l)?	Broad suppression of lymphocytes and myeloid cells Possibly faster acting than rituximab alone Cyclophosphamide sparing in severe renal AAV (RITUXVAS[59])	No data on comparison with rituximab alone No reduction in adverse events compared with standard cyclophosphamide	Suggested benefits include a glucocorticoid-sparing effect Often also used in patients with refractory disease
Should PLEX be used to improve overall survival and the rate of kidney failure?	Greater early kidney recovery Reduced risk of ESKD at 1 year (meta-analysis)[64] Renal benefits are greatest in patients with highest baseline creatinine	No demonstrated effect on mortality Benefits for ESKD risk attenuate after 1 year Costly and invasive procedure Increased risk of infection Dosing is unclear, and was not aimed at ANCA negativity in clinical trials	Possibly only short-term benefits Consider for individuals who have creatinine levels of >300 µmol/l Imlifidase might enable more complete ANCA removal
Is a tailored rituximab regimen based on changes in ANCA levels and B cell numbers a convenient strategy to maintain remission?	Lower cumulative rituximab exposure compared with a fixed-schedule regimen Possible avoidance of unnecessary treatment in patients with a lower risk of relapse Possible lower risk of acquired immunodeficiency	Higher rates of relapse compared with a fixed-schedule regimen Inconsistent association between changes in ANCA levels and B cell numbers and relapse (particularly in GPA and PR3-AAV) Need for closer monitoring and more frequent visits to detect changes	Expectant approach could be acceptable in patients at a lower risk of relapse and milder disease or renal-limited AAV with ESKD More definitive treatments needed (next-generation anti-CD20 monoclonal antibodies, BiTE and CAR T cell therapy)

BiTE, bispecific T cell engager; CAR, chimeric antigen receptor; ESKD, end-stage kidney disease; GPA, granulomatosis with polyangiitis; GTI, glucocorticoid toxicity index; MPL, methylprednisolone.

at 6 months than those who continued on 5 mg per day (15.5% versus 4.2%, $P = 0.022$). However, no difference was observed among patients who received rituximab[69].

Both the MAINRITSAN trial, which enrolled newly diagnosed patients induced with cyclophosphamide, and the RITAZAREM trial, which recruited patients with relapsing AAV who received rituximab for induction, showed the superiority of rituximab over azathioprine for maintenance of remission, with a comparable safety profile[70,71] (Table 2). The dose of rituximab varied in these studies from 0.5 g every 6 months to 1 g every 4 months for 2 years, respectively. Relapse rates after completion of 2 years of therapy were similar in the rituximab and azathioprine arms of the RITAZAREM trial, despite a high cumulative dose, which indicates that the effect of these treatments wears off[71]. Thus, there is no evidence that prolonged treatment with rituximab 'resets' the immune system in patients with relapsing AAV and prevents the return of autoimmune manifestations, thereby modifying the long-term disease course. As a result, repeat dosing is often needed and extended therapy beyond 2 years results in fewer relapses, but the optimal duration of treatment is not established[72,73]; however, rituximab increases the risk of acquired immunodeficiency, which causes hypogammaglobulinaemia, reduces response to vaccines and infections, and tailored regimens have been used to limit exposure[74]. In a randomized clinical trial, administering rituximab upon an increase in ANCA levels or B cell repopulation was associated with a lower cumulative rituximab dose than a fixed-schedule regimen, but with a numerically higher risk of relapse, which raises questions about the ability of these biomarkers to guide individual therapy[75].

Persistent ANCA positivity or reappearance after induction therapy is associated with a higher risk of relapse than stable ANCA negativity, and might warrant continuation of rituximab. However, only half of the patients who are ANCA-positive experience a relapse, whereas episodes of minor relapse can occur without changes in ANCA level or in patients who are ANCA negative[52,68,73,76–78]. Clinical phenotype could facilitate the interpretation of serial ANCA measurements, as an increase in ANCA levels seems to be more strongly associated with relapse in patients with vasculitic manifestations, particularly kidney involvement, than in those with non-renal and granulomatous disease[78]. Similarly, MPO-ANCA status shows a higher association with risk of relapse than PR3-ANCA, possibly because it is more strongly associated with vasculitic manifestations. A study of a large cohort of patients with MPO-AAV and kidney involvement found that relapses are virtually absent in patients who achieve and maintain MPO-ANCA negativity, whereas the risk is markedly increased with reoccurrence or persistence of MPO-ANCA[77]. Hence, rituximab could be discontinued in patients with renal-limited involvement who achieve MPO-ANCA negativity, given the low likelihood of relapse, and be resumed in the case of seroconversion, which shows good predictability, but decisions are more challenging for patients with PR3-ANCA and extra-renal disease.

Although a reappearance of B cell number indicates that the effects of rituximab are waning, the association of B cell repopulation with relapse is variable. Some studies have reported that B cell reconstitution by 12–18 months preceded all episodes of relapse, whereas in other cohorts relapse occurred regardless of B cell counts[52,75,79,80]. A single-centre, randomized trial compared rituximab given upon B cell repopulation with that given after an increase in the level of ANCAs in patients who had completed 24 months of fixed-schedule rituximab[81]. The former led to fewer relapses but was associated with a higher rituximab exposure (average 1 g per year), which confirms the limitations of tailored approaches and the short-term effects of rituximab.

An advantage of the rituximab fixed interval approach is the low rate of relapse and the predictability and probable reduction in monitoring visits; however, the cumulative dose and the risk of infection and secondary immunodeficiency with rituximab could be higher than with a tailored regimen[74,82]. Using an expectant approach, whereby further rituximab treatment is only given when disease returns or an increase in ANCA levels occurs, requires close monitoring and good communication with the managing centre (Table 3). The consequences of a potential relapse should be considered based on patient prior disease course. If the effects of a relapse are likely to be mild, such as constitutional symptoms and ENT symptoms only, then an expectant approach is acceptable, whereas if major complications have occurred previously, such as loss of kidney function, then this approach would increase the risk of substantial organ damage, such as ESKD.

Advances in B cell-targeting and T cell-targeting therapies

The efficacy of rituximab supports the concept that B cells have a pivotal role in AAV[32,59,71,83,84]. B cells are precursors of plasmablasts and plasma cells, which are short-lived and long-lived antibody-producing cells, respectively, and can present antigens to T cells and orchestrate the immune response by secreting cytokines. Notably, rituximab does not target the progenitor cells that reconstitute the B cell pool, plasmablasts or plasma cells, all of which are CD20-negative and could account for persistent or recurrent ANCA positivity that underlies refractory and relapsing disease[52]. Furthermore, although complete B cell depletion is usually observed in the peripheral blood following rituximab treatment, memory B cells persist in lymphoid and target organs and have been associated with active disease[85–88]. B cell repopulation following rituximab is more frequently delayed in AAV than in other diseases, and has been associated with inherent defects in B cell development that predate the start of immunosuppression, which raises concerns about the toxicity of this therapy[89].

T cells are involved in the pathogenesis of AAV[90]. Serum markers of T cell activation, such as soluble IL-2 receptor, are elevated in patients with AAV and correlate with disease activity, whereas the expression of transcriptomic profiles of T cell exhaustion is associated with prolonged remission[91,92]. T helper (T_H) cells support B cell responses to ANCA antigens and the production of ANCAs, which are high-affinity, class-switched IgG1 antibodies. Furthermore, T_H cells are expanded in active disease and secrete cytokines that are associated with a T_H1 (IL-12 and IFNγ) and T_H17 profile (IL-6 and IL-17), which promote inflammation and damage[93,94].

Emerging B cell-depleting and B cell-inhibiting therapies developed for lymphoproliferative disorders and autoimmune diseases and agents that affect T cells have been studied in AAV. The following paragraphs discuss B cell-targeting and T cell-targeting treatments of interest for AAV (Table 4 presents the main characteristics of these therapies and Fig. 1 depicts their targets).

B cell-depleting therapies

The next-generation anti-CD20 monoclonal antibodies show greater B cell-depleting capacity and better tolerability than rituximab[95,96]. Ofatumumab, a fully human monoclonal antibody that elicits increased complement-dependent cytotoxicity, was used in seven patients with AAV, some of whom were intolerant or unresponsive to rituximab (Table 4). The treatment was effective and well tolerated, but no trial has been conducted with this agent[97]. Obinutuzumab is a humanized monoclonal antibody that causes increased antibody-dependent cytotoxicity and direct cell death via a glycoengineered Fc and has been used to successfully treat a small number of patients[98] (Fig. 1). The ongoing ObiVas trial (ISRCTN13069630) is testing the hypothesis that obinutuzumab is superior to rituximab at inducing tissue B cell depletion in patients with PR3-AAV, the primary end point being the relative percentage change from baseline in the number of $CD19^+$ cells in the nasal associated lymphoid tissue at week 26 (ref. 99). The ObiVas trial is not powered to compare the clinical efficacy of obinutuzumab with rituximab, but results could provide biological evidence of its superiority. Deeper B cell depletion might translate into a more complete and durable remission, which would be particularly beneficial for patients with refractory disease and would reduce the need for retreatment compared with rituximab. Obinutuzumab has shown promising results in lupus nephritis and membranous nephropathy, in which response to rituximab is often incomplete, but its effect in AAV will need to be determined[100,101].

Anti-CD19 monoclonal antibodies, such as inebilizumab, hold promise, because they can deplete CD20-negative plasmablasts, but they have not yet been used in AAV[102,103]. Daratumumab, an anti-CD38 monoclonal antibody that primarily targets plasma cells, seemed to be successful in a few patients with AAV who were refractory to rituximab and other treatments, which suggests that targeting the broader B cell lineage improves remission rates in AAV[104–106] (Fig. 1). The combination of an anti-CD20 and an anti-CD38 therapy has proved beneficial in some autoimmune diseases and is being evaluated in paediatric nephrotic syndrome (NCT05704400)[107,108]. Notably, CD38 is also expressed on myeloid cells and some B cell subsets; hence, the beneficial effects of daratumumab might also come about, because this therapy targets these cells in addition to plasma cells.

B cell-inhibiting therapies

Targeting molecules involved in B cell development and maturation, such as B cell activating factor (BAFF, which is also known as BLyS), could inhibit pathogenic responses without B cell depletion[109]. BAFF promotes the survival of autoreactive B cells and levels of BAFF are increased in patients with AAV compared with healthy individuals[110–114] (Fig. 1). The BREVAS trial assessed the addition of monthly intravenous belimumab, a human monoclonal antibody that neutralizes soluble BAFF and is licensed for systemic lupus erythematosus (SLE), to azathioprine for remission maintenance in GPA and MPA[115] (Table 4). Recruitment in this trial was terminated early owing to a shift towards rituximab as a maintenance agent and no effect of belimumab on relapse rate was observed; however, among the 14 patients in the belimumab arm who received rituximab at induction, no relapse occurred, suggesting that combining the two agents could have additive effects. The randomized, placebo-controlled COMBIVAS trial is evaluating the addition of subcutaneous belimumab (200 mg weekly from day 1 for 52 weeks) to a standard induction regimen of rituximab (1,000 mg on day 8 and 22) in PR3-AAV[116]. This small study (36 patients) has a primary mechanistic end point of time to PR3-ANCA negativity. Exploratory endpoints were

Table 4 | Characteristics of B cell-targeting and T cell-targeting therapies under evaluation in antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis (AAV)

Therapy	Target	Features	Mechanism	Strengths	Main study	Comments
B cell-depleting agents						
Obinutuzumab	CD20	Humanized IgG1 monoclonal antibody with a glycoengineered Fc	B cell depletion via enhanced ADCC and DCD Internalized less than rituximab	Improved tissue B cell depletion compared with rituximab	ObiVas (ISRCTN13069630)[99]	Potential for more complete and durable remission than rituximab
Ofatumumab	CD20	Fully human IgG1 monoclonal antibody	B cell depletion, enhanced CDC and slower dissociation than rituximab	Improved tolerability compared with rituximab	McAdoo et al. (case series)[97]	Encouraging results in patients who are intolerant or unresponsive to rituximab
Daratumumab	CD38	Humanized IgG1 monoclonal antibody	Depletion of plasmablasts and plasma cells, also targets T cells, memory B cells and NK cells	Depletion of autoantibody-producing cells	Ostendorf et al. (case report)[105] and Rixecker et al. (case report)[106]	Effective in patients who were refractory to rituximab
CAR T cells	CD19	Engineered autologous T cells	Virtually complete depletion of B cells and plasmablasts	Eradication of autoreactive B cell clones and rapid repopulation with immature, non-autoreactive B cells	Mouse model[124] and several trials in humans (NCT06152172, NCT06590545, NCT06508346)	Potential for long-term drug-free remission High costs and limited safety data
B cell-inhibiting agents						
Belimumab	BAFF	Fully human IgG1 monoclonal antibody	Inhibition of B cell survival and maturation	Preferential inhibition of autoreactive B cells	BREVAS[115]	Fewer relapses only in patients treated with belimumab and rituximab at induction
Combination of rituximab and belimumab	CD20 and BAFF	Chimeric IgG1 and fully human IgG1	Combined B cell depletion and inhibition	Increased mobilization of memory B cells and block of post-rituximab surge in BAFF	COMBIVAS[116]	Potential for enhanced response to rituximab and longer time to relapse
Telitacicept	BAFF and APRIL	TACI-Fc fusion protein	Inhibition of B cell differentiation and plasma cell functions	Potent inhibition of autoantibody production	TTCAAVREM (NCT05962840)	Findings limited to Chinese studies of patients with other rheumatic diseases (such as SLE)
Povetacicept	BAFF and APRIL	TACI-Fc fusion protein	Superior inhibition of B cell differentiation and plasma-cell functions than telitacicept	Potent inhibition of autoantibody production	RUBY-3 (NCT05732402)	Open-label study on various immune-mediated kidney diseases
T cell-targeting agents						
Alemtuzumab	CD52	Humanized IgG1 monoclonal antibody	Prolonged depletion of T cells and, to a lesser extent, B cells, NK cells and other innate immune cells	Expansion of tolerogenic T regulatory cells at reconstitution	ALEVIATE[132]	Low rate of complete response in patients with refractory AAV and high frequency of early relapses
Abatacept	CD80	CTLA4-Fc fusion protein	Inhibition of co-stimulatory signals of B cells and other APCs	Inhibition of T cell activation	Langford et al.[133] ABROGATE[134]	Preliminary results show no effect on the rate of treatment failure in relapsing, non-severe GPA
Ustekinumab	p40 subunit of IL-12 and IL-23	Humanized IgG1 monoclonal antibody	Inhibition of T helper 1 and T helper 17 responses	Reduced neutrophil priming and recruitment and reduced macrophage activation	Engesser et al.[135]	Supported by transcriptome data of kidney tissue from patients with AAV and glomerulonephritis

ADCC, antibody-mediated cytotoxicity; APCs, antigen-presenting cells; APRIL, A proliferation-inducing ligand; BAFF, B cell activating factor of the tumour necrosis factor family; CDC, complement-dependent cytotoxicity; CTLA4, cytotoxic, T lymphocyte-associated protein 4; DCD, direct cell death; GPA, granulomatosis with polyangiitis; NK, natural killer; SLE, systemic lupus erythematosus, TACI, transmembrane activator and calcium modulator and cyclophilin ligand interactor.

designed to assess the effect of the rituximab–belimumab combination therapy within both peripheral blood and lymphoid tissues. Observations that BAFF protects B cells from rituximab-induced depletion and promotes their reconstitution suggests that belimumab could enhance response to B cell-depleting therapy and consolidate remission of AAV[117]; however, this combination therapy has shown conflicting results in SLE[118,119].

A proliferation-inducing ligand (APRIL) is another cytokine of the TNF family that promotes B cell functions and shares receptors with BAFF, which includes transmembrane activator and calcium modulator and cyclophilin ligand interactor (TACI). Dysregulation of the BAFF–APRIL system has been implicated in AAV and trials of telitacicept (NCT05962840) and povetacicept (NCT05732402), dual BAFF and APRIL inhibitors (that are based on TACI), have been launched in GPA and MPA[120] (Fig. 1).

Cell-based therapies

Anti-CD19 chimeric antigen receptor (CAR) T cells are engineered autologous lymphocytes that can eliminate CD19$^+$ cells[121] (Fig. 1). This therapy induces a profound depletion of B cells and plasmablasts and has achieved a sustained clinical and serological drug-free remission in patients with autoimmune diseases refractory to conventional treatment[122,123]. Anti-CD19 CAR T cells protected mice from MPO-ANCA-induced glomerulonephritis and also attained ANCA negativity and complete B cell depletion in the bone marrow of a patient with refractory AAV[124,125]. Several phase I–II studies will evaluate CAR T cell therapy in patients with AAV.

Despite the potential of anti-CD19 CAR T cells providing a cure by eradicating autoreactive cells, implementing this therapy in AAV remains challenging (Table 4). The production of CAR T cells is expensive and involves several steps, including leukapheresis, lentivirus-based transduction and in vitro expansion. Furthermore, patients need to receive a lymphodepleting conditioning regimen before reinfusion and be on low doses of glucocorticoids to enable good in vivo cell expansion. Therefore, the treatment requires clinical stability and might not be compatible with remission induction regimens for acute presentations of AAV. A more suitable population to treat might be patients with non-severe, refractory or frequently relapsing disease who need longer-term drug-free complete remission, thus perhaps justifying the high costs of the treatment. The risks of the known adverse events of CAR T therapy in AAV, including cytokine release syndrome and immune effector cell-associated neurotoxicity syndrome, are not known but could also alter the risk-to-benefit ratio and influence patient selection.

The use of allogeneic CAR T cells from healthy donors, which, unlike autologous ones, can be immediately available, has been proposed, but these cells might cause graft-versus-host disease and be rapidly rejected by the recipient[126]. Furthermore, T cells or natural

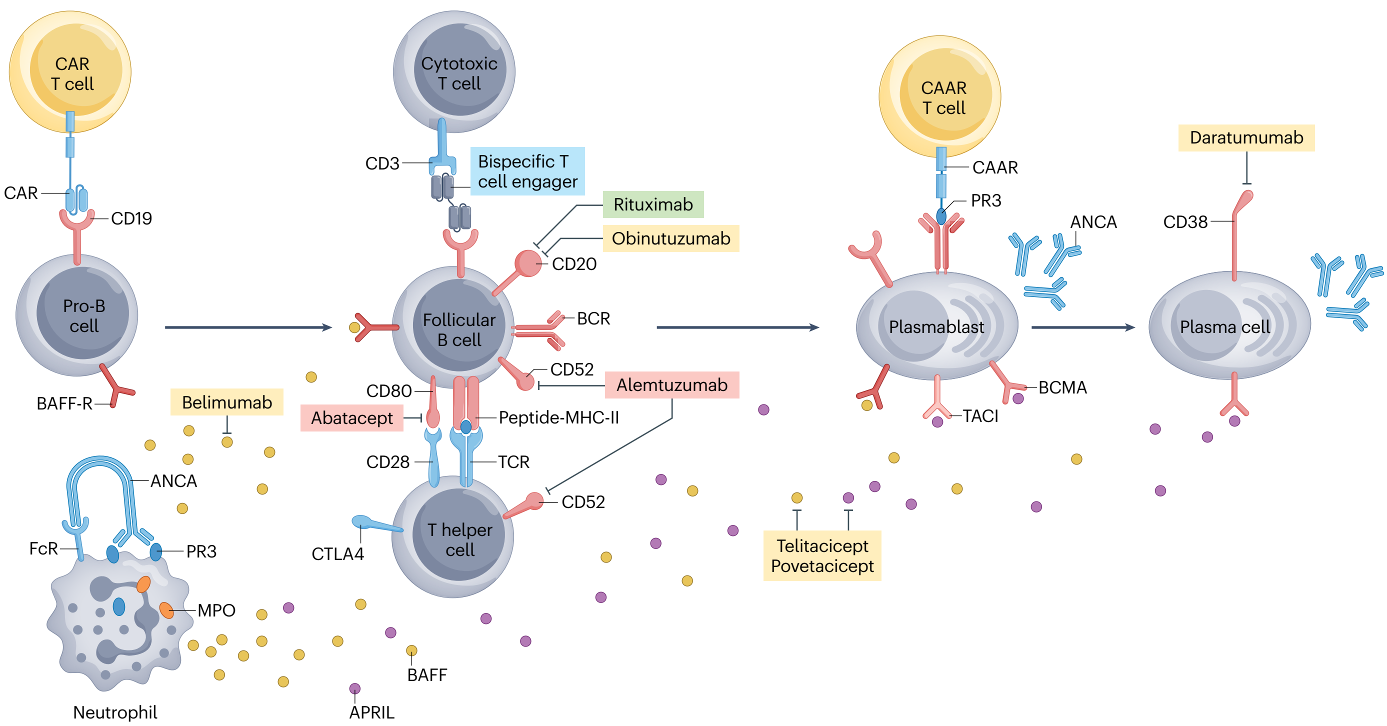

Fig. 1 | B cell-targeting and T cell-targeting therapies for AAV. Therapies that are currently approved for antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis (AAV) are shown in green; therapies in red are those that do not have a major effect in AAV; therapies highlighted in yellow are currently under investigation or show positive results in case reports; and therapies in blue have been proposed for AAV but thus far have only been used for other conditions. CD19 is expressed by pro-B cells, B cells and plasmablasts, which can be depleted by chimeric antigen receptor (CAR) T cells or by bispecific T cell engagers. Immature, mature and memory B cells (not shown) express CD20 and are depleted by rituximab, obinutuzumab and ofatumumab (not shown), whereas plasma cells are targeted by daratumumab, which binds to CD38. Autoreactive B cells that recognize molecules such as proteinase 3 (PR3) via their B cell receptor (BCR) could be selectively eliminated by chimeric autoantigen receptor (CAAR) T cells with PR3 as the autoantigen. B cells and T cells express CD52, which is the target of alemtuzumab. Follicular B cells provide co-stimulatory signals to T helper cells and can be inhibited by abatacept, a CTLA4-like fusion protein. ANCA-stimulated neutrophils produce B cell-activating factor of the TNF family (BAFF), a pro-survival B cell factor, which is antagonized by belimumab. The B cell-stimulating cytokines BAFF and APRIL share receptors, such as TACI; TACI-like fusion proteins, including telitacicept and povetacicept, can neutralize both ligands. Other BAFF receptors include BAFF-R and BCMA, expressed at various stages of the B cell lineage. APRIL, A proliferation-inducing ligand; BAFF-R, BAFF receptor; BCMA, B cell maturation antigen; CTLA4, Cytotoxic T-Lymphocyte Antigen 4; MPO, myeloperoxidase; TACI, transmembrane activator and calcium modulator and cyclophilin ligand interactor.

killer cells engineered to express a chimeric autoantigen receptor (CAAR) have been studied in some autoantibody-mediated conditions and might be used in AAV[127,128] (Fig. 1). CAAR cells are recognized by autoreactive cells and can selectively eradicate them, thereby sparing the rest of the B cell pool. Another option is bispecific T cell engagers, antibodies that target a B cell marker, such as CD19, and CD3 on T cells to create an immunological synapse and enhance killing of B cells, which has been used in rheumatoid arthritis[129] (Fig. 1). The field of cell-based therapy is rapidly evolving and several approaches to in vivo cell production are being explored.

T cell-targeting therapies

T cells are affected by several therapies used in AAV, including glucocorticoids, cyclophosphamide, azathioprine and mycophenolate mofetil. Furthermore, cyclosporine A, a calcineurin inhibitor that suppresses activation and proliferation of T cells, showed positive results as a maintenance agent in a small trial of patients with GPA[130]. An analysis of the RITUXVAS trial found that the presence of T cell tubulitis was associated with a poor response to treatment with rituximab, highlighting the clinical relevance of targeting T cells in therapy[131]. Emerging T cell-targeted therapies have been evaluated in AAV.

Alemtuzumab is an anti-CD52 monoclonal antibody licensed for multiple sclerosis that induces sustained depletion of T cells, and to a lesser extent of B cells, monocytes and eosinophils (Fig. 1). The phase IIb ALEVIATE trial included 12 patients with AAV who were refractory to standard therapy, and compared 30 mg with 60 mg of alemtuzumab. Complete remission was only achieved in one third of patients and relapses were frequent; hence, alemtuzumab has not been proposed as an alternative to rituximab[132].

Abatacept, a fusion protein comprising an inert Fc portion and the extracellular domain of the cytotoxic T lymphocyte-associated protein 4 (CTLA-4), binds to the costimulatory proteins CD80 and CD86 and prevents T cell activation (Fig. 1). Despite good results in a pilot study[133], the phase III ABROGATE trial (NCT02108860), which compared abatacept with placebo in relapsing, non-severe GPA, demonstrated no difference in the rate of treatment failure, arguing against major benefits of this therapy in AAV[134].

Evidence of enrichment of $CD4^+$ T cells that produce T_H1 and T_H17 cytokines in the kidneys of patients with renal AAV has prompted consideration of ustekinumab, a human monoclonal antibody targeting the p40 subunit common to both IL-12 and IL-23, which promote the differentiation of T_H1 and T_H17 cells respectively (Fig. 2). Use of ustekinumab in combination with glucocorticoids and low-dose cyclophosphamide was associated with good clinical response in four patients with relapsing AAV[135].

Advances in complement-targeting therapies

The complement system comprises the classical, lectin and alternative pathways, which converge into the generation of complement component 5 (C5), which is cleaved into the anaphylatoxin C5a and the membrane attack complex, also known as C5b-9. The role of complement in AAV has long been considered minor, although complement breakdown products are detected in the serum and tissues of several patients, and correlate with disease activity[136–139]. In the past two decades, animal models have shown that the complement system, particularly the alternative complement pathway, is crucial in the development of vasculitic lesions[140–142]. C5a was identified as a critical primer of neutrophils and monocytes for ANCA-induced activation, which results in further C5a generation, sustaining the amplification of the process (Fig. 2). Genetic deletion of C5a and pharmacological inhibition of the C5a receptor (C5aR1, also known as CD88) prevents the formation of glomerular crescents in a mouse model of MPO-ANCA-induced glomerulonephritis and led to the development of therapies that target the C5a–C5aR1 axis[142,143]. The following sections discuss complement inhibition in AAV.

Avacopan

Avacopan (formally known as CCX168), an orally administered small molecule, inhibits C5aR1 but does not interfere with C5b-9, which has an important role in the defence from encapsulated bacteria, such as *Neisseria meningitidis* (Fig. 2). Its approval in 2021 as a first-in-class complement inhibitor was a milestone in the field of AAV, because avacopan is the first therapy developed with GPA and MPA as the primary indication. The phase II trials CLEAR and CLASSIC, and the phase III trial ADVOCATE showed efficacy and good tolerability of avacopan as an adjunctive therapy and its potential as a glucocorticoid-sparing therapy[33,144,145].

ADVOCATE, a double-blind, double-dummy trial, compared avacopan 30 mg twice daily for 52 weeks with prednisone 60 mg per day, tapered to discontinuation at 21 weeks, in combination with cyclophosphamide followed by azathioprine or rituximab without azathioprine, among 330 patients with newly diagnosed or relapsing AAV[33]. Participants could receive open-label intravenous methylprednisolone and oral prednisone for up to 4 weeks. Avacopan was non-inferior to prednisone in achieving remission at week 26 and superior in sustaining remission up to the end of the study (52 weeks). The avacopan group received two-thirds lower mean cumulative glucocorticoid dose than the prednisone group and suffered less glucocorticoid toxicity. Furthermore, patients on avacopan reported greater improvements in health-related quality of life than those on prednisone, which might reflect better disease control and a reduction in some of the adverse effects of glucocorticoids, such as insomnia, weight increase and skin fragility[146].

Avacopan was also associated with earlier reductions in proteinuria and greater kidney function recovery, with the best improvements observed in those patients with lower baseline eGFR (15–30 ml/min/1.73 m^2). The increase in eGFR continued beyond week 26 with avacopan, but not with prednisone, indicating sustained beneficial effects[147].

Current guidelines recommend considering avacopan as an alternative to the extensive use of glucocorticoids, in addition to cyclophosphamide or rituximab, in patients with active disease, particularly in those at an increased risk of glucocorticoid toxicity and in those with severe kidney involvement[36]. With use of avacopan, which is immediately active, rapid withdrawal of glucocorticoids at 4 weeks can be achieved for most patients. Furthermore, although avacopan has efficacy in maintaining remission, whether it has additive benefits for patients receiving rituximab for maintenance and whether it should be continued beyond 1 year remains unknown. The inhibition of neutrophil activation and degranulation by avacopan, which results in reduced release of MPO and PR3, might lower the antigenic drive that promotes the immune response towards ANCA antigens, thereby contributing to relapse prevention.

Currently, the high costs of avacopan limits both access to this therapy and the duration of treatment, and long-term treatment outcomes are not available. Post-authorization studies, such as a phase IV randomized trial (NCT06072482) and a prospective registry (NCT05897684), will further evaluate the safety of avacopan and clarify whether treatment translates into improved rates of kidney failure and overall survival[148]. Reassuringly, avacopan seemed beneficial in

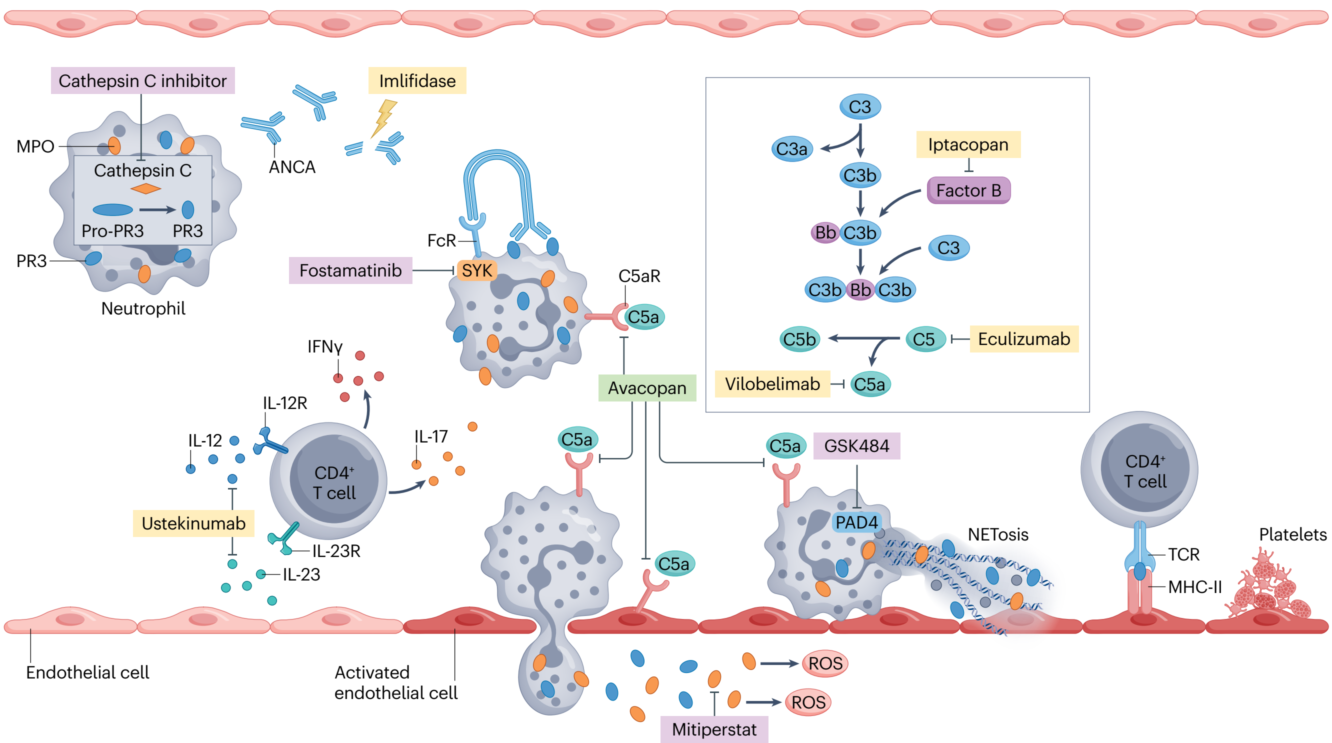

Fig. 2 | Innovative therapies targeting mechanisms of ANCA-induced inflammation. Highlighted in green are therapies currently approved for antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis; highlighted in yellow are therapies that have been used or are being tested in patients with ANCA-associated vasculitis; and therapies highlighted in purple show benefits in vitro or in other conditions. Cathepsin C inhibitors might prevent the activation of proteinase 3 (PR3) in the lysosomes of immature neutrophils. ANCAs can be degraded by imlifidase, ANCA-induced activation of neutrophils and monocytes (not shown) through engagement of the Fc receptor (FcR) and subsequent SYK signalling can be inhibited by fostamatinib. The alternative complement pathway, which leads to the generation of C5a, is crucial for neutrophil priming, chemotaxis and degranulation, as well as endothelium activation. Complement activation can be targeted at different stages by iptacopan (factor B), eculizumab (C5), vilobelimab (C5a) and avacopan (C5aR). Activated neutrophils release PR3 and myeloperoxidase (MPO), which enhances reactive oxygen species (ROS) production and subsequent endothelial inflammation. This process can be inhibited by mitiperstat. Neutrophils also undergo NETosis, extruding decondensed chromatin and toxic proteins, such as MPO and PR3, which are recognized by T cells. GSK484 inhibits PAD4, a key enzyme for neutrophil extracellular trap (NET) formation. IL-12 and IL-23 promote T_H1 and T_H17 cell differentiation, respectively, which results in the release of pro-inflammatory T_H1 and T_H17 cytokines, such as IFNγ and IL-17, and these pathways can be inhibited by ustekinumab. Endothelial cells become activated in response to inflammation and acquire a pro-thrombotic phenotype, which facilitates platelet aggregation. PAD4, peptidylarginine deiminase 4.

real-world populations, including in patients with hypoxic alveolar haemorrhage and eGFR <15 ml/min/1.73 m^2 or those requiring dialysis, who were excluded from the ADVOCATE trial, suggesting an effect across a wider range of presentations[149–151].

Other complement-targeting therapies

Vilobelimab (IFX-1) neutralizes C5a and prevents binding to C5aR1 and also C5L2, the other receptor for C5a, which might serve as negative modulator of the C5aR signalling pathway[152] (Fig. 2). Two phase II trials, IXPLORE (NCT03712345) and IXCHANGE (NCT03895801), showed the safety and efficacy of vilobelimab compared with prednisone, with a lower glucocorticoid cumulative dose, but further studies are not currently planned[153].

Eculizumab, and the longer-acting ravulizumab, inhibit C5 and prevent the generation of C5a and also C5b-9, which has a central role in the pathogenesis of complement-mediated thrombotic microangiopathy (Fig. 2). A few patients with AAV, including those complicated by thrombotic microangiopathy, have been successfully treated with eculizumab[154–156]. Unlike avacopan, anti-C5 therapies require vaccination and antibiotic prophylaxis against encapsulated bacteria.

Targeting components of the alternative pathway proximal to C5, such as C3, factor B and factor D, is supported by observations that their breakdown products correlate with disease activity in AAV[137,157–159]. Iptacopan, an oral inhibitor of factor B, will be investigated in a phase II study in AAV (NCT06388941) (Fig. 2); in addition, factor H-related proteins, which inhibit the main regulator of the alternative pathway, are increased in AAV and have been proposed as a therapeutic target[157]. Other emerging agents that target C3 and other aspects of the complement pathway are under consideration for treatment of autoimmune diseases, including AAV.

Although deposition of C3 and factor B in the kidney and reduced serum levels of C3 have been consistently associated with a more severe disease phenotype, no study has assessed how these factors affect response to complement antagonists[137,139,158]. Patients with signs of complement activation might benefit more from therapies that target the complement pathway and future work could identify markers that could guide the use of these therapies.

Novel targets and biomarkers

Several therapies that could complement immunosuppressive treatment in AAV are being developed. An improved understanding of disease pathogenesis has enabled the identification of novel therapeutic targets to prevent inflammation induced by ANCA. Furthermore, there is increasing awareness that fibrosis and other comorbidities, such as cardiovascular disease and immunodeficiency, contribute substantially to damage in patients with AAV and should be specifically treated. Biomarkers that enable early identification of disease and anticipate relapse could also improve management. The following section discusses advances in novel targets and biomarkers (Table 5 presents features of these therapies and Fig. 2 illustrates some of the drugs and their targets.)

Targeting antineutrophil cytoplasmic antibody-induced inflammation

Imlifidase, a recombinant protease derived from *Streptococcus pyogenes*, degrades circulating and tissue-bound human IgG and rapidly cleared ANCAs in patients with anti-GBM disease who were double-seropositive for anti-GBM antibodies and ANCA in the GOOD-IdeS-01 trial[160] (Fig. 2). Successful use of imlifidase in a patient with PR3-AAV and refractory lung haemorrhage has been reported[161]. The phase II study INFLIMIDARDSe (EudraCT-Nr. 2021-004706-22) will evaluate imlifidase in patients with AAV-related alveolar haemorrhage (Table 5).

PR3 requires activation by the lysosomal protease cathepsin C during neutrophil maturation. In vitro, cathepsin C inhibition diminishes neutrophil cell-surface PR3 expression, resulting in reduced capacity for PR3-ANCA to induce cell activation, NET formation and glomerular endothelial cell injury[162,163] (Fig. 2). These findings might prompt further investigation of cathepsin C inhibitors in AAV, which are under evaluation for bronchiectasis[164] (Table 5). MPO secreted by activated neutrophils mediates oxidative injury and correlates with kidney disease severity[165]. Inhibition of MPO ameliorated kidney inflammation in a mouse model of crescentic glomerulonephritis[166]. Mitiperstat (AZM198), an oral MPO inhibitor, has been studied in heart failure and could be considered for the treatment of AAV[167] (Fig. 2).

Spleen tyrosine kinase (SYK) mediates intracellular signalling of various receptors, including Fc engagement by ANCAs, and is upregulated in ANCA-activated neutrophils and monocytes[168] (Table 5). Increased expression of SYK is noted in glomeruli of patients with active

Table 5 | Potential strategies for the management of patients with antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis (AAV)

Agent	Type	Mechanism	Effect	Main study
ANCA-induced inflammation				
Imlifidase	Recombinant IgG protease	Cleavage of IgG	Rapid clearance of ANCA	INFLIMIDARDSe (EudraCT-Nr. 2021-004706-22)
BI-9740	Oral cathepsin C inhibitor	Reduced expression of PR3	Reduced PR3-ANCA-induced neutrophil activation and NET formation	Jerke et al. (in vitro study)[163]
Niclosamide	Oral anthelminthic	Inhibition of the IL-6– STAT3 pathway and other pathways	Reduced PR3-stimulated formation of granuloma	Henderson et al. (in vitro study)[209] and Lim et al.[210]
Mitiperstat (AZM198)	Oral MPO inhibitor	Inhibition of ROS generation and NET formation	Reduced neutrophil degranulation and oxidative injury	Antonelou et al. (mouse model)[166]
Fostamatinib	Oral SYK inhibitor	Inhibition of SYK activation through ANCA engagement of Fc receptors	Reduced neutrophil and monocyte activation	McAdoo et al. (mouse model)[170]
GSK484	Oral PAD4-inhibitor	Inhibition of NET formation	Reduced MPO tissue deposition and immune response to MPO	O'Sullivan et al. (mouse model)[174]
Vilobelimab	Anti-C5a monoclonal antibody	Inhibition of C5a-mediated chemotaxis and priming of neutrophils	Inhibition of ANCA-induced neutrophil activation and glucocorticoid-sparing	IXPLORE IXCHANGE[153]
Other pro-inflammatory pathways				
Iptacopan	Oral Factor B inhibitor	Inhibition of the alternative complement pathway	Reduced complement-mediated neutrophil chemotaxis and priming	NCT06388941
Hydroxychloroquine	Oral antimalarial	Inhibition of lysosomal activity and signalling of TLR7 and TLR9	Reduced secretion of pro-inflammatory cytokines	HAVEN (NCT04316494)[211]
Fibrosis				
Lixudebart (ALE. F02)	Anti-claudin-1 monoclonal antibody	Targeting epitopes of exposed, non-junctional claudin-1	Reduced activation of pro-fibrogenic pathways	RENAL F02 (2022-502184-38)
Pirfenidone	Oral anti-fibrotic agent	Inhibition of TGFβ, PDGF, MMP and anti-inflammatory effects	Reduced fibroblast activity and chronic inflammation	NCT03385668
Cardiovascular disease				
Sparsentan	Endothelin A and angiotensin II receptor antagonist	Reduced vasoconstriction and pro-thrombotic effects	Improved endothelial dysfunction, fibrinolysis and nephroprotection	SPARVASC (NCT05630612)

ANCA, antineutrophil cytoplasmic antibody; MMP, matrix metalloproteinase; MPO, myeloperoxidase; NET, neutrophil extracellular trap; PAD4, peptidylarginine deiminase 4; PDGF, platelet-derived growth factor; PR3, proteinase 3; ROS, reactive oxygen species; STAT, signal transducer and activator of transcription; SYK, spleen tyrosine kinase; TGFβ, transforming growth factor β; TLR, Toll-like receptor.

Glossary

Alternative complement pathway
This pathway does not require a trigger, such as immune complexes, but is instead constitutively activated through the 'tick over' of C3 and regulatory factors, such as factor H, which are required to prevent widespread activation in the absence of pathogenic stimuli.

Anaphylatoxin
A cytokine that mediates the recruitment and activation of innate immune cells.

Chimeric antigen receptor
(CAR). An engineered receptor that contains an antibody fragment that recognizes unprocessed antigen, such as CD19, and transmembrane and intracellular signalling domains of the T cell receptor complex, which provide activation and co-stimulatory signals in response to antigen binding.

Chimeric autoantigen receptor
(CAAR). An engineered receptor that contains a portion of an autoantigen, potentially PR3, which can be recognized by autoreactive cells, and transmembrane and intracellular domains of the T cell receptor complex.

Cytokine release syndrome
A supraphysiologic inflammatory response (which can progress to multiorgan failure) that can occur following treatment with any immune therapy that activates or engages endogenous or infused T cells and/or other immune effector cells.

Granulomatous inflammation
A form of tissue inflammation that is characterized by the presence of granulomas, structures in which palisading macrophages, including multinucleated giant cells and epithelioid cells and lymphocytes aggregate to surround a core of material, such as apoptotic neutrophils, that cannot be eliminated.

Immune effector cell-associated neurotoxicity syndrome
Neurological symptoms (ranging from mild confusion to seizures and cerebral oedema) that can occur after treatment with any immune therapies that activate or engage endogenous or infused T cells and/or other immune effector cells, leading to damage of the blood–brain barrier.

Immune checkpoints
Molecules that provide co-stimulatory or co-inhibitory signals to T cells that can support or inhibit their activation.

Membrane attack complex
A structure that occurs when the components of the terminal complement pathway assemble and form pores in the cell membrane that can lead to cell lysis.

Neutrophil extracellular traps
(NETs). Net-like structures composed of decondensed chromatin and granule proteins, such as MPO and PR3, that are released as a form of programmed cell death by neutrophils.

T cell tubulitis
Inflammation in the tubule-interstitial compartment of the kidney predominantly driven by T cells.

renal AAV[169]. Fostamatinib, an oral SYK inhibitor licensed for immune thrombocytopenia, ameliorated both kidney and lung lesions in a mouse model of MPO-AAV[170] (Fig. 2).

ANCAs also induce NETs, which are toxic to the endothelium and stimulate further ANCA responses[171]. Peptidylarginine deiminase 4 (PAD4) citrullinates histones and is essential for NET formation[172,173] (Fig. 2). GSK484, an oral inhibitor of PAD4, abrogated NET production, MPO deposition and glomerular inflammation in a mouse model of MPO ANCA-induced glomerulonephritis, and might have a role in the treatment of AAV[174] (Table 5).

Fibrosis

Claudin-1, a component of the epithelial tight junctions that is overexpressed and exposed at non-junctional sites by parietal epithelial cells in glomerular crescents, has been associated with glomerulosclerosis (fibrosis within the glomeruli)[175,176]. Lixudebart (ALE.01), a monoclonal antibody that targets an epitope of the exposed, non-junctional claudin-1, reduced proteinuria and fibrosis in a mouse model of MPO-ANCA-induced glomerulonephritis[177]. The phase II trial RENAL-F02 (NCT06047171) is investigating whether lixudebart slows the progression to CKD in patients with renal AAV (Table 5).

Interstitial lung disease (ILD) is common in patients who are MPO-ANCA positive, and portends a poor prognosis[178,179]. Anti-fibrotic therapies, such as nintedanib and pirfenidone, are now approved for idiopathic pulmonary fibrosis, but evaluation in MPO-ANCA-ILD is limited to retrospective series[180,181]. A pilot study on pirfenidone in MPO-ANCA-ILD (NCT03385668) has completed recruitment and the results are awaited (Table 5).

Biomarkers

In the past decade, urine biomarkers with the potential to improve detection of a renal flare have been evaluated (Supplementary Table 1). Levels of urinary soluble CD163 (usCD163), a macrophage-shed scavenger receptor, are increased in AAV-related glomerulonephritis and can differentiate patients with active renal vasculitis from healthy individuals and those in remission[182,183]. A large multicentre analysis defined a usCD163–creatinine ratio threshold of 250 ng/mmol as a cut-off for active renal vasculitis and a validated diagnostic kit is now commercially available[184]. Notably, usCD163 levels reflect glomerular inflammation but are not specific to AAV and are also being evaluated in other types of glomerulonephritis[185].

Further potential blood and urinary biomarkers of disease activity have been proposed and might be combined to improve accuracy in detecting and monitoring disease flares (Supplementary Table 1). In the RAVE trial, levels of circulating immune checkpoints at study enrolment were associated with failure of rituximab and with both sustained remission and infections[186]. If these findings are replicated in independent cohorts, the use of these markers might facilitate personalized immunosuppressive treatment.

Preventing infections

During the COVID-19 pandemic, the negative effects of immunosuppressive therapies, particularly rituximab, on the development of protective antibodies following vaccination were highlighted[187,188]. In patients treated with rituximab, T cell responses do not seem to be affected, whereas B cell depletion predicted failure to achieve seroconversion[189,190]. Humoral response could be increased by additional booster vaccine doses and even with undetectable anti-spike antibodies, vaccinated individuals had a five-fold reduced risk of moderate or severe COVID-19, compared with unvaccinated individuals[191]. The PNEUMOVAS trial compared a single versus double or quadruple dose of conjugated pneumococcal vaccines (PCV13) at the start of treatment with rituximab, prior to a dose of pneumococcal polysaccharide vaccine (PPV23), in 95 patients with active AAV. Preliminary results showed that no strategy achieved response to all serotypes but when stratified for age, patients on the double dose of PCV13 had a higher probability of response than those on the single or quadruple dose[192]. ACQUIVAS (NCT03514979) is an open-label, phase IIb study also assessing response to single doses of PCV13 and PPV23 in patients with AAV in remission. Immunogenicity of those treated with rituximab and those who received other regimens will be compared.

Chronic kidney and cardiovascular disease

Cardiovascular morbidity and mortality are elevated in patients with AAV, and the risk is particularly high among those with CKD[28,193–195]. Sodium glucose cotransporter-2 inhibitors are now a mainstay in the treatment of CKD and are used in patients with AAV, although trials leading to approval of these drugs either excluded (the DAPA-CKD trial) or only included a small number of patients with AAV (the EMPA-KIDNEY trial)[196,197]. Patients with AAV-related glomerulonephritis have also been excluded from trials of mineralocorticoid agonists[198]. Exclusion of this high-risk group contributes to care inequity and therapeutic disenfranchisement.

An investigator-initiated study found that plasma endothelin-1 is increased in patients with AAV in remission, compared with healthy volunteers, and correlates with arterial stiffness and endothelial dysfunction, which are known to increase the risk of cardiovascular events. Endothelin-1 inhibition was able to attenuate these changes and sparsentan, a dual endothelin-1 receptor A and angiotensin II receptor antagonist, is being evaluated (NCT05630612) in patients with AAV[199].

Conclusions

Current treatments for AAV achieve high rates of survival and remission but patients remain burdened with a substantial risk of relapse and toxicity. Emerging therapies provide promising opportunities to further improve outcomes. The excellent results of rituximab encourage the evaluation of next-generation anti-CD20 monoclonal antibodies, whereas use of B cell-inhibiting therapies, such as belimumab, and T cell-targeting agents, such as abatacept, seems less successful. Furthermore, recognition of the importance of the complement system in the pathogenesis of AAV has led to the approval of avacopan, which holds promise in achieving a substantial reduction in glucocorticoid dose, with better disease control and greater kidney recovery. A major unmet need is the development of a safe, long-term treatment for AAV; cell-based therapies are rapidly evolving and might offer a cure to patients with AAV, although translation of these therapies into clinical practice is challenging. Furthermore, pathways involved in ANCA-induced inflammation and fibrosis have been identified as potential targets. Improving trial endpoints will also be crucial to show the superiority of emerging therapies to those currently available. In this regard, alongside large global trials, studies on smaller and homogeneous populations can focus on mechanistic endpoints and biomarkers. The management of AAV has made major steps forward in the past decades and continues to advance steadily.

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Author contributions

G.T., M.C.M., A.K. and S.M. researched data for the article. G.T., A.K., R.M.S., R.B.J., P.A.M. and D.J. contributed substantially to the discussion of the content. G.T., M.C.M., A.K., S.P.M. and D.R.W.J. wrote the article. G.T., R.M.S., B.T., S.M., R.B.J., P.A.M. and D.R.W.J. reviewed and/or edited the manuscript before submission.

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